INTERRUPTS FOR BEGINNERS · ARM GICV3 / V4

GIC from Zero

You do not need to know anything about chips to start. Each level adds one small idea, first as an everyday story, then as the real chip word. By the end you will understand how an Arm processor's Generic Interrupt Controller (GIC) works and how engineers test it.

1 · What is an interrupt
2 · Why a controller
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Word list

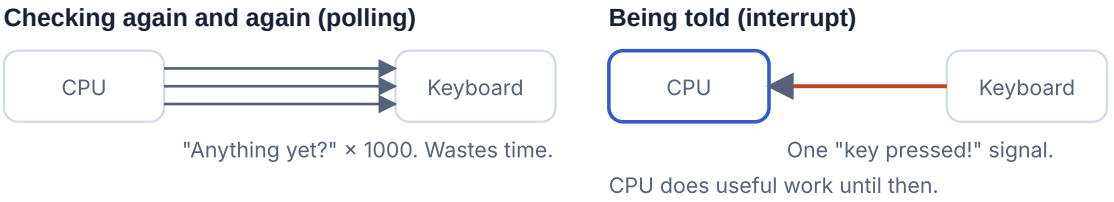
1

THE VERY FIRST IDEA

What is an interrupt?

Story.

You are doing homework and waiting for a friend. You could get up every two minutes to check the door. Or you could keep working and let the **doorbell** tell you when they arrive. The doorbell is an interrupt.



Every keyboard press, network packet, timer tick and disk finish arrives at the CPU as an interrupt.

THE REAL WORDS

An **interrupt** is a signal that says "stop what you are doing for a moment, something needs you". The CPU pauses its current program, runs a small piece of code called the **interrupt handler**, then goes back to where it was.

Why are interrupts better than checking again and again?

- The CPU can do other work until it is needed
- Interrupts make the keyboard faster
- The CPU never has to run any code

2

WHY WE NEED A HELPER

Why a controller?

Story.

A big school has hundreds of bells, phones and alarms, and many classrooms. If every bell rang in every classroom, nobody could study. So the school has a **front office**. Every bell rings at the office first. The office decides *which classroom* should hear it and *how urgent* it is.

> THE REAL WORDS
>
> A modern chip has many **cores** (small CPUs, each one is called a **PE**, processing element) and hundreds or thousands of interrupt sources. The **Generic Interrupt Controller (GIC)** is the front office. Arm wrote one rule book for it, the **GIC architecture specification**, so every Arm chip's GIC behaves the same way for software.

Remember: The GIC collects interrupts, picks the right core and the most urgent one, and tells that core.

3

DIFFERENT KINDS OF BELLS

The four kinds of interrupt

Every interrupt has a number called its **INTID** (interrupt ID). The number tells you which kind it is.

SGI · a note from a classmate

`INTID 0 – 15`

Software Generated Interrupt. One core sends a message to another core: "please wake up" or "please refresh your list".

PPI · your own alarm clock

`INTID 16 – 31`

Private Peripheral Interrupt. Belongs to one core only, like that core's own timer. Every core has its own copy with the same number.

SPI · the school phone

`INTID 32 – 1019`

Shared Peripheral Interrupt. Comes from a shared device (UART, USB, DMA). The GIC can send it to any core you choose.

LPI · a letter in the post

`INTID 8192 and up`

Locality-specific Peripheral Interrupt. Not a wire, a *message* written to memory by devices like PCIe cards. There can be thousands.

Numbers 1020–1023 are special "answers", not real interrupts. 1023 means "nothing for you right now". Newer GICs also have extra PPI and SPI number ranges (1056–1119 and 4096–5119).

A core's own timer fires. Which kind is it?

SGI | PPI | SPI | LPI

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HOW THE SIGNAL LOOKS

A doorbell press or a light left on

Edge: a short press

ding!



The GIC remembers "it rang once".

Level: a light left on

on until the device is served



Stays on until software fixes the cause.

Each interrupt is set as edge or level in the GIC's configuration register (ICFGR).

THE REAL WORDS

Edge-triggered: the GIC notices the moment the signal goes up. **Level-sensitive**: the interrupt is "on" for as long as the wire is high. With level, the handler must make the device lower the wire, or the interrupt comes straight back.

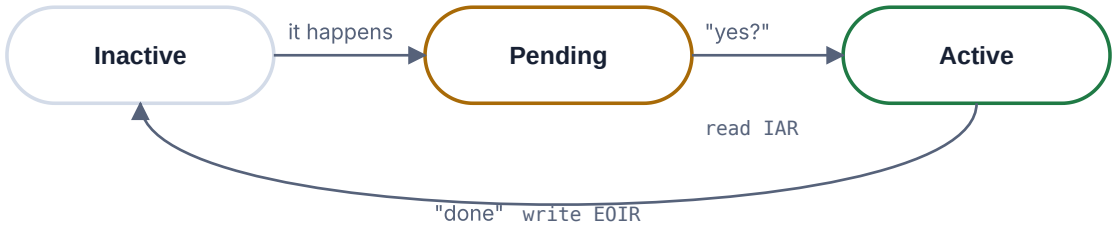
5

HAND UP, ANSWERED, DONE

The life of one interrupt

Story.

A student raises a hand (**pending**). The teacher says "yes?" (**acknowledge**), and now the teacher is busy with that student (**active**). When the question is answered, the teacher says "done" (**end of interrupt**) and the student puts the hand down (**inactive**).



If it happens again while Active, it becomes "Active and Pending" and is served once more afterwards.

THE REAL WORDS

The CPU reads a register called `ICC_IAR1_EL1` (**Interrupt Acknowledge Register**). The value it reads is the INTID it should handle. When it is finished, it writes that INTID to `ICC_EOIR1_EL1` (**End Of Interrupt Register**). Message interrupts (LPIs) are simpler: they only have Inactive and Pending, so reading IAR finishes their state change.

The CPU has read IAR but has not written EOIR yet. What state is the interrupt in?

Inactive | Pending | Active

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FIRE ALARM BEATS LUNCH BELL

Who goes first: priority

Story.

If the fire alarm and the lunch bell ring together, you go with the fire alarm. And if you are in the middle of lunch and the fire alarm rings, you stop eating. But if a friend calls while you are handling a fire, they wait.



THE REAL WORDS

Each interrupt has an 8-bit **priority**. **A smaller number means more urgent.** The core has a **Priority Mask Register** (`ICC_PMR_EL1`): only interrupts with a number *smaller* than the mask can reach it. While the core is handling one interrupt, its **running priority** is that interrupt's priority, and only something more urgent can **preempt** (interrupt the interrupt).

PMR is 0xA0. A lunch bell with priority 0xC0 rings. What happens?

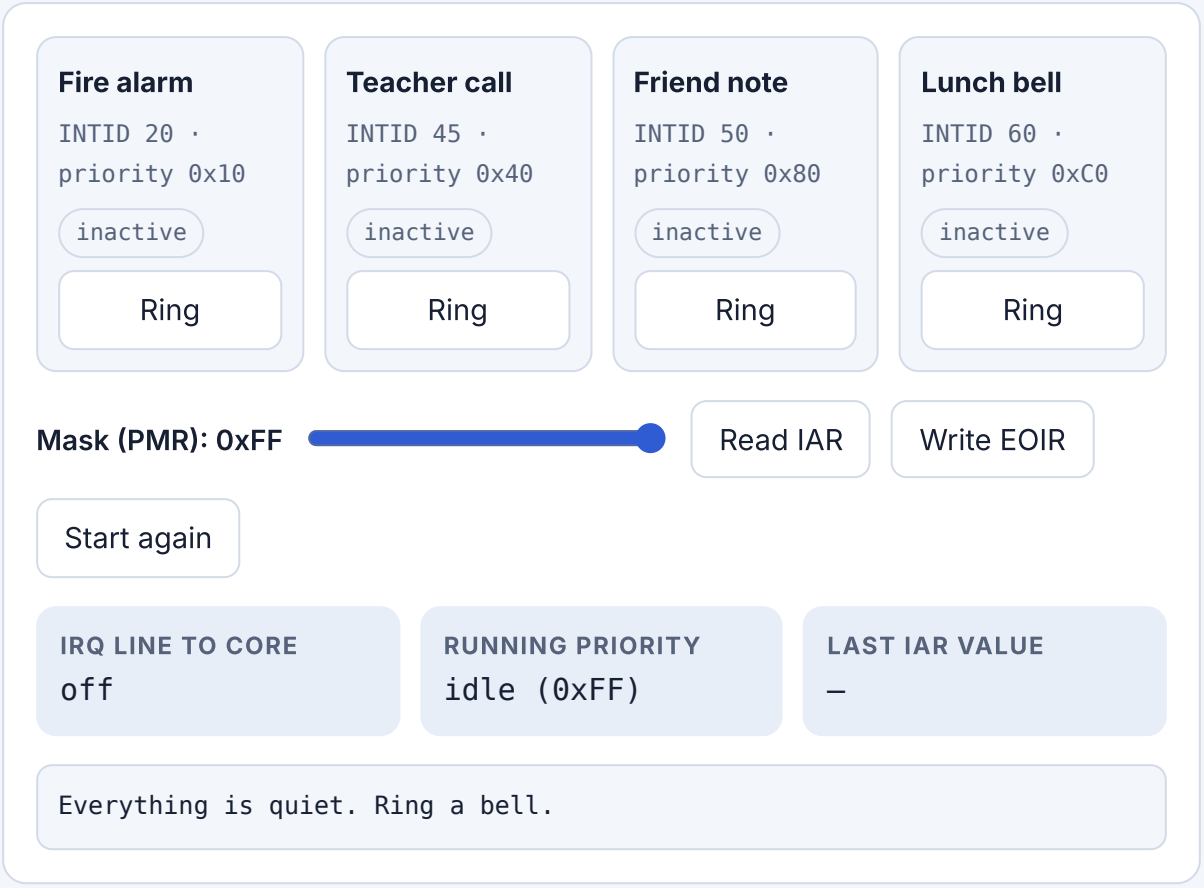
- The core takes it
- It waits, because 0xC0 is not smaller than 0xA0
- It is deleted

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PLAY WITH IT

Try it: be the CPU

Ring some interrupts, move the mask, then press **Read IAR** to take the most urgent one and **Write EOIR** to finish it. Watch the IRQ light: it turns on only when something is allowed through.

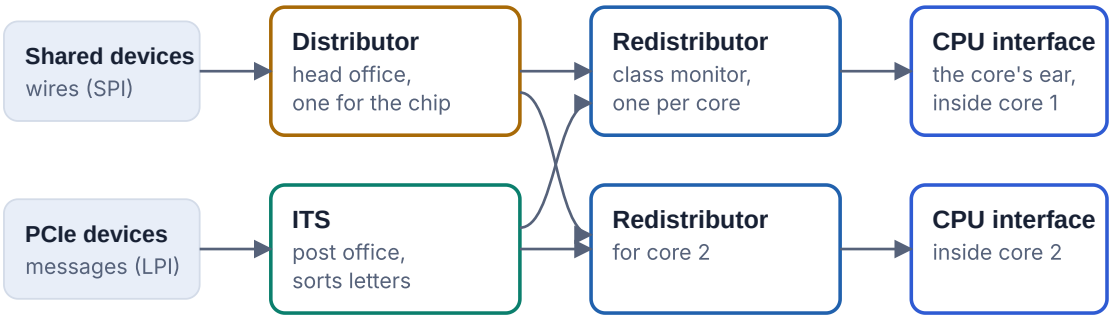


Things to try: ring Lunch then Fire and read IAR (Fire wins). Read IAR on Teacher, then ring Fire (it preempts, IRQ lights). Set the mask to 0x30 and ring Teacher (IRQ stays off, IAR returns 1023).

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THE OFFICE HAS DEPARTMENTS

The four helpers inside the GIC



SGIs and each core's own PPIs start at the Redistributor. Messages between Redistributor and CPU interface travel on a small link called the GIC Stream Protocol.

HELPER	SCHOOL VERSION	WHAT IT REALLY DOES	REGISTER NAMES START WITH
Distributor	Head office	Holds settings for shared interrupts (SPIs): on/off, priority, which core.	`GICD_…`
Redistributor	Class monitor	One per core. Holds SGI, PPI and LPI settings for that core and picks its most urgent pending interrupt.	`GICR_…`
CPU interface	Your ear	Inside the core. Applies the mask and priority rules, raises the IRQ or FIQ signal, gives you the INTID.	`ICC_…`
ITS	Post office	Turns device messages into LPIs and sends them to the right Redistributor.	`GITS_…`

How many Redistributors does a chip with 8 cores have?

1 | 8, one per core | It depends on the number of SPIs

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FINDING THE RIGHT CLASSROOM

Every core has an address

Story.

To deliver a note in a big school you need four parts: campus, building, floor, room. "Campus 0, building 1, floor 2, room 3" points to exactly one room.

THE REAL WORDS

Each core has an **affinity** address with four numbers: `Aff3.Aff2.Aff1.Aff0`, for example `0.0.1.3`. The Distributor has one `GICD_IROUTER` register per SPI that stores the address of the core that should get it. You can also set a mode that means "any one free core", which spreads work around.

To send an SGI, a core writes the target addresses into `ICC_SGI1R_EL1`. It can also say "everyone except me".

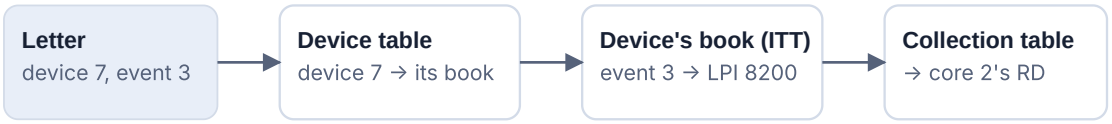
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LETTERS, NOT WIRES

The post office: LPIs and the ITS

Story.

A letter arrives with a sender name (which device) and a letter number (which event). The post office has address books. It looks up the sender, then the letter number, and finds out which room it must go to and what to call it.



Software fills these tables with commands in a queue: MAPD (add a device), MAPTI (add an event), MAPC (point a group of interrupts at a core).

THE REAL WORDS

A device writes to the ITS register `GITS_TRANSLATER`. The ITS uses the **DeviceID** and **EventID** to find the LPI number and the target Redistributor. LPI settings (priority, enable) are kept in a table in normal memory, so there can be many thousands of LPIs without huge chip area.

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THE PRINCIPAL'S OFFICE

Secret rooms: groups and security

Story.

Some messages are for the principal only, some for the teachers, and some for everyone. Normal students cannot even read the principal's mail.

GROUP	SCHOOL VERSION	WHO HANDLES IT
Group 0	Principal's own mail	The most trusted firmware (EL3). Always arrives as **FIQ**.
Secure Group 1	Locked staff room	The secure world (for example a trusted OS that keeps keys safe).
Non-secure Group 1	Regular classroom	The normal OS, such as Linux or Android.

THE REAL WORDS

Arm cores have two "doorbell" signals: **IRQ** and **FIQ**. Simple rule: an interrupt for the world the core is in right now arrives as IRQ. An interrupt for a different world arrives as FIQ, which goes to the firmware so it can switch worlds. Normal software cannot see or change secure interrupts.

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MAKE-BELIEVE CLASSROOMS

Pretend computers: virtualization

Story.

One big classroom is split into pretend classrooms. A helper (the **hypervisor**) takes real messages and puts **sticky notes** on the right pretend desk. Later, the school gets a smarter post office that can stick notes on pretend desks by itself, without bothering the helper.

THE REAL WORDS

A **virtual machine (VM)** is a pretend computer running on real cores. In GICv3 the hypervisor writes **virtual interrupts** into **list registers** (`ICH_LR<n>_EL2`), the sticky notes. The VM reads them as if they were real. In **GICv4**, the ITS can deliver message interrupts straight to a VM (**direct injection**). If that VM is not running right now, a **doorbell** interrupt tells the hypervisor to schedule it.

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SAVING BATTERY

Sleeping cores

Story.

Before a student naps, they tell the class monitor "I'm going to sleep". The monitor says "OK, I'll hold your messages". If something important comes, the monitor wakes them up.

THE REAL WORDS

Software sets `ProcessorSleep` in the `GICR_WAKER` register, then waits until `ChildrenAsleep` shows the CPU interface is quiet. Then power can be turned off. An interrupt for that core now creates a **wake request** to the power controller instead of an IRQ.

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BEING A DETECTIVE

How engineers test a GIC

Design verification (DV) engineers make sure the chip follows the rule book before it is built, using simulation. Their job is to imagine what could go wrong and prove it does not.

The basic recipe

1. **Stimulus**: make interrupts happen (wiggle wires, send messages, write registers).
2. **Reference model**: a simple program that knows the rules and predicts the right answer.
3. **Checker** (scoreboard): compares what the chip did with the prediction.
4. **Coverage**: a checklist proving every kind of situation was tried.

Detective questions to ask

- What if two interrupts ring at exactly the same time?
- What if the mask changes while the IRQ light is on?
- What if a level light turns off just after "yes?"
- What if a core goes to sleep while a message is on the way?
- What if the post office's address book changes while a letter is being sorted?
- Can normal software peek at a secret interrupt?

Remember: **Good testers check what software can see (the IAR value, the IRQ line, register values), not tiny internal timing.**

What is a reference model for?

It makes the chip run faster

It predicts the correct behaviour so the chip can be checked against it

It replaces the real chip

A–Z

KEEP THIS HANDY

Word list

WORD	SIMPLE MEANING
PE / core	One small CPU inside the chip.
INTID	The number of an interrupt.
Pending	Hand raised, waiting.
Active	Being handled right now.
IAR	Register the CPU reads to say "yes?" and learn the INTID.
EOI / EOIR	Register the CPU writes to say "done".
Priority	How urgent. Smaller number = more urgent.
PMR	Do-not-disturb line for one core.
Preempt	A more urgent interrupt interrupts the current one.
Edge / level	Doorbell press / light left on.
Affinity	A core's four-part address.
ITS	Post office that turns device messages into LPIs.
IRQ / FIQ	The two interrupt signals into an Arm core.
Hypervisor / VM	Helper that runs pretend computers / a pretend computer.
List register	Sticky note holding a virtual interrupt for a VM.
Spurious (1023)	"Nothing for you right now."

Ready for the next step?

Once these 14 levels feel easy, move to the **GIC Field Guide**, which covers the same ideas at senior-engineer depth: register details, ITS commands, GICv4, corner-case bugs and interview questions.